Comparative Time Series Forecasting of Oil Prices

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Github Link: https:/github.com/thulluri-ramya/Comparative-Time-Series-Forecasting-of-Oil-Prices

1. Introduction :

Oil prices have a major impact on the global economy because they affect transportation costs, energy output, and overall economic activity. As a result, the ability to accurately estimate oil price changes is valuable to governments, investors, and enterprises making strategic and financial decisions. However, forecasting oil prices is challenging since they are influenced by a wide range of unknown factors, including geopolitical events, supply and demand movements, and global market fluctuations.
Time series forecasting methods are widely used to analyze historical data and anticipate future results. Traditional statistical methods, such as Autoregressive Moving Average (ARMA) and Autoregressive Integrated Moving Average (ARIMA) models, are frequently used to evaluate time series data[2].

Recently, machine learning and deep learning techniques have been increasingly used to handle forecasting problems. Long Short-Term Memory (LSTM) networks have shown greater performance when dealing with sequential data[1]. LSTM models can learn intricate patterns and long-term correlations from time series data[1], making them excellent for modeling financial and economic variables.

The purpose of this study is to examine and forecast oil price trends using statistical and deep learning techniques. In particular, an ARIMA model and an LSTM neural network are developed and used to oil price data. The performance of these models is quantified using standard forecasting error metrics, and both models are used to create forecasts for the coming 24 months.The results are then compared to understand the strengths and limitations of each modelling approach.

2. Data Description and Pre-Processing:

The dataset used in this analysis contains historical observations of oil prices measured in US dollars over time. Each observation corresponds to a specific date and the associated price of oil on that date. Because the data is structured chronologically, it is suitable for time series analysis and forecasting.

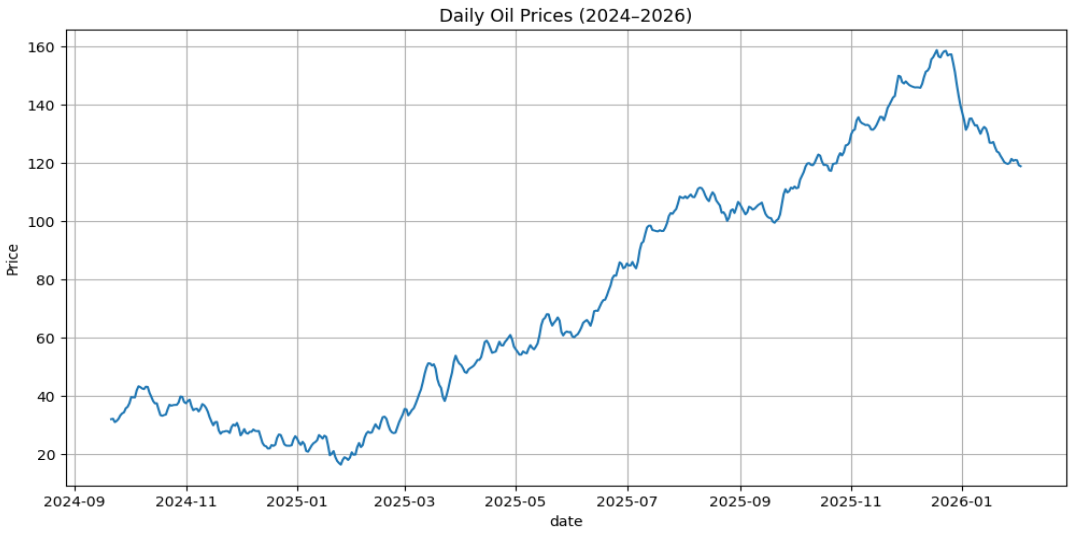

Figure 1: Historical oil price time series used for modelling.

Several preprocessing steps were taken to prepare the dataset for forecasting models. First, the date column was converted to datetime format and used as the dataset's index. This step is crucial because it allows the data to be accurately processed as a time series. A visual study of the oil price series was then conducted by plotting the data over time. This helped to discover general patterns including trends, variations, and periods of high volatility. Basic descriptive statistics were also computed to help us understand the dataset's characteristics. These figures show the mean, standard deviation, minimum, and maximum value of oil prices. Such summary metrics provide insight into the data's distribution and variability.

Table 1: Summary Statistics

count 500.000000	mean 75.873980
std 42.912398	min 16.480000
25% 35.372500	50% 64.290000
75% 111.572500	max 158.780000

For the LSTM model, an additional preprocessing step was required. Neural network models generally perform better when the input values are scaled to a similar range. Therefore, the oil price values were normalised using a MinMaxScaler, which transformed the data to a range between 0 and 1. This scaling improves the stability and efficiency of the model during the training process.

3. Stationarity Analysis:

Stationarity is a requirement for several classic time series designs, including ARIMA[2] as an example. A stationary time series maintains its statistical properties across time, and this includes a constant mean and variance. If a time series is not stationary, the outcomes of the model may become unreliable.

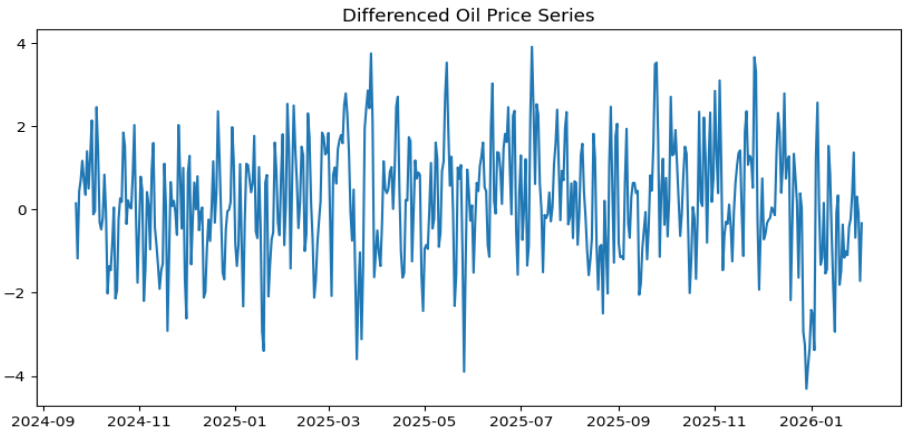

Figure 2: Differenced oil price series used to achieve stationarity.

The Augmented Dickey-Fuller (ADF) test was performed to determine whether the oil price data were stationary. The null hypothesis for the ADF test asserts that the time series is non-stationary due to the presence of a unit root. The data is considered non-stationary, and the null hypothesis cannot be rejected if the final p-value exceeds 0.05.

The results of the ADF test showed that the original oil price series was not stationary. This outcome is common for financial time series, as they often contain trends and changing levels over time. To address this issue, first-order differencing was applied to the data. Differencing works by subtracting earlier observations from the current one, which helps remove the trend and stabilise the series.

After applying differencing, the stationarity of the data was examined again. The transformed series showed more stable behaviour and was suitable for modelling using the ARIMA approach. In addition, Autocorrelation Function (ACF) and Partial Auto Correlation Function (PACF) plots were created to help choose optimal model parameters.

4. ARIMA Model:

The ARIMA model is a popular statistical tool for analyzing and forecasting time series data[2]. The model consists of three components: autoregressive (AR), differencing (I), and moving average (MA). The autoregressive component represents the link between an observation and its previous values, whereas the moving average component represents the relationship between an observation and past prediction mistakes. The differencing component is employed to make the series stationary when necessary.

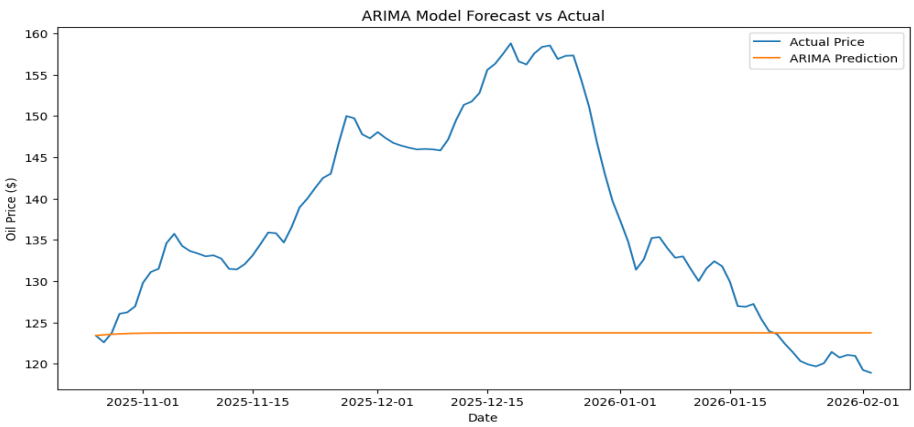

Figure 3: ARIMA predictions compared with actual oil prices.

In this study, the ARIMA model was applied to the oil price dataset after confirming the stationarity of the differenced series. The ACF and PACF plots were examined to identify appropriate values for the autoregressive and moving average parameters. Based on this analysis, a suitable combination of parameters was selected for the model.

To discover the best ARIMA configuration, a grid search approach was employed to explore various parameter combinations. The parameters p, d, and q denote the autoregressive, differencing, and moving average orders, respectively. In this work, the parameter space was explored using p values from 0 to 8, d values from 0 to 2, and q values from 0 to 8. The Akaike Information Criterion (AIC) was calculated for each possible combination. The model with the lowest AIC value was chosen as the ideal configuration because it provides a better balance of model fit and complexity.

These predictions were compared with the actual observed oil prices to assess the accuracy of the model.

Table 2: ARIMA Evaluation Metrics

MAE	14.572486526233384
RMSE	18.047123733676408

Model performance was measured using two assessment metrics: mean absolute error (MAE) and root mean squared error (RMSE). These metrics indicate the average difference between the projected values and the data. Lower values for these indicators indicate higher forecasting accuracy.

The trained ARIMA model was also used to generate forecasts for the next 24 months. Confidence intervals were included to represent the uncertainty associated with these future predictions.

Residual Analysis:

After the ARIMA model was fitted, residual diagnostics were performed to check whether it had adequately represented the structure of the data. The residuals should ideally resemble white noise, with a roughly normal distribution and no noticeable autocorrelation. To search for residual correlations in the model errors, an autocorrelation function (ACF) plot of the residuals was generated. The model effectively captured the majority of the series' temporal structure, as indicated by the absence of significant spikes in the residual ACF. To determine whether the errors were nearly normally distributed, a histogram was employed to assess the residuals' distribution.

5. LSTM Model:

In addition to the statistical ARIMA model, an LSTM neural network was used to forecast oil prices. LSTM networks are recurrent neural networks that are specifically built to handle sequential data[1]. Unlike standard neural networks, LSTM models may store information from earlier time steps, allowing them to detect long-term dependencies in time series data.

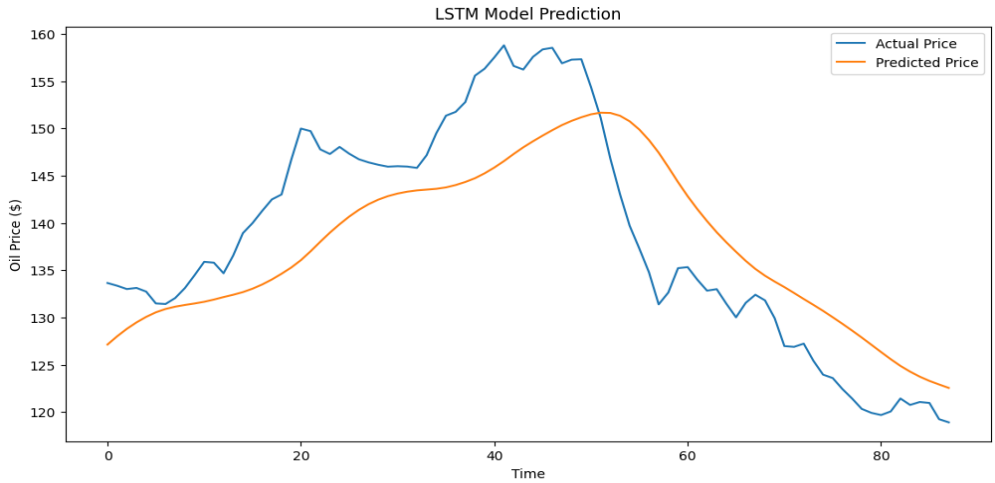

Figure 4: LSTM model predictions compared with actual values.

Before training the LSTM model, the oil price values were scaled using a MinMaxScaler to normalise the data between 0 and 1 The model's input sequences were then created using a sliding window technique. To anticipate the next value in the sequence, this method used a fixed number of previous observations (known as the window size).

The LSTM architecture employed in this work was composed of two LSTM layers followed by a dense output layer. The Adam optimisation technique was used to train the model, with the loss function set to mean squared error. During training, the model discovered patterns and linkages in historical oil price data.

After training, the model was utilized to make predictions for the test dataset. The anticipated values were then converted back to their original scale and compared to real oil prices. Similar to the ARIMA model, MAE and RMSE were used to evaluate the performance of the LSTM model.

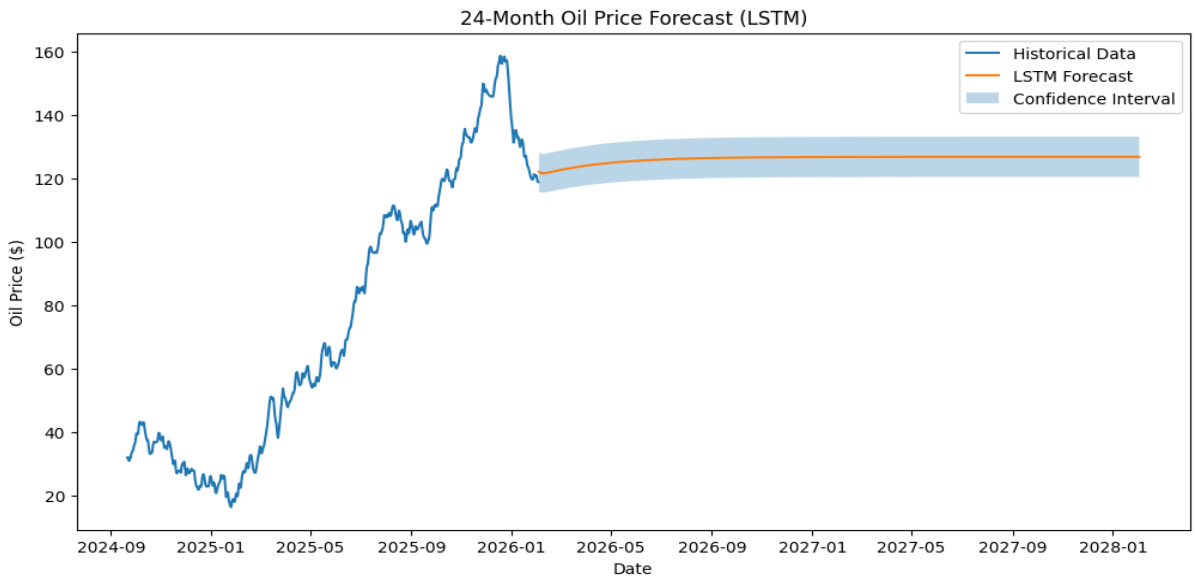

Figure 5: 24-month oil price forecast generated using the LSTM model.

Finally, the trained LSTM model was used to forecast oil prices for the next 24 months, allowing for a comparison between the statistical and deep learning approaches.

6. Results and Model Comparison:

After implementing both the ARIMA and LSTM models, their forecasting performance was evaluated using standard error metrics. The models were tested using the test portion of the dataset, and their predictions were compared with the actual observed oil prices.

Table 3: comparison table from both model Results.

Model	**MAE**	**RMSE**
ARIMA	14.57	18.05
LSTM	6.41	7.23

Two evaluation criteria were employed to assess the forecasts' accuracy: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). While RMSE gives greater weight to higher errors and indicates the overall forecasting accuracy, MAE computes the average absolute difference between anticipated and actual values.

The LSTM model outperformed the ARIMA model in terms of error levels, as can be seen from the results. This suggests that the patterns in the oil price time series were better captured by the LSTM model. The LSTM model's capacity to identify intricate and nonlinear correlations in sequential data may be responsible for its enhanced performance.

In addition to numerical evaluation, the forecasting performance was also assessed visually by comparing the predicted values with the actual observations. The ARIMA model was able to capture the overall trend of the oil price series but sometimes struggled to follow sudden fluctuations. In contrast, the LSTM model produced predictions that more closely followed the variations in the data.

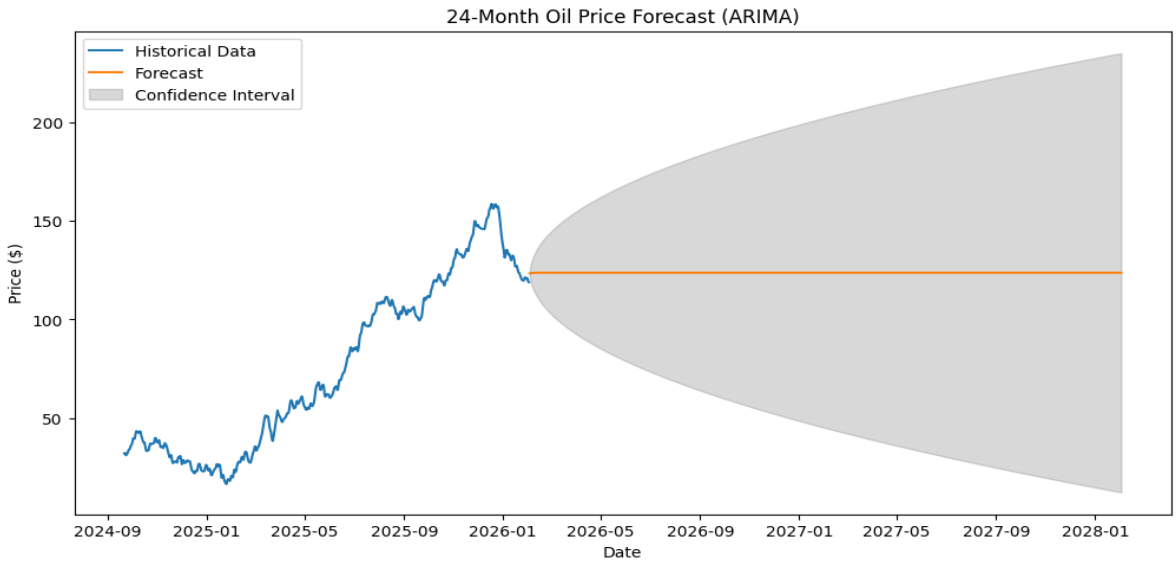

Figure 6: 24-month oil price forecast generated using the ARIMA model.

Both models were also used to generate forecasts for the next 24 months. The ARIMA forecast included confidence intervals, which illustrate the uncertainty associated with long-term predictions. As expected, the width of the confidence intervals increased over time, reflecting the growing uncertainty in forecasting further into the future.

When comparing the predicted values with the actual oil prices in the testing dataset, it was observed that the LSTM model followed the fluctuations in the series more closely than the ARIMA model. While the ARIMA model was able to capture the general trend of the time series, it struggled to react quickly to sudden changes in oil prices. In contrast, the LSTM model was better able to adapt to short-term variations, resulting in lower forecasting errors.

7. Discussion and Critical Analysis:

Although both the ARIMA and LSTM models were able to generate reasonable forecasts, their performance differed due to the nature of their modelling approaches.

The ARIMA model is based on linear relationships within the time series data. While it performs well when patterns are stable and the data is stationary, it may struggle to capture complex nonlinear behaviour. Oil prices are influenced by many unpredictable factors[3], such as geopolitical events, global demand changes, and economic policies. Because ARIMA relies primarily on historical patterns, it may not fully capture these dynamic influences.

On the other hand, the LSTM model is capable of learning nonlinear relationships and long-term dependencies within the data. This allows the model to adapt more effectively to the fluctuations commonly observed in financial time series. The improved accuracy of the LSTM model in this study suggests that deep learning approaches may be more suitable for modelling complex economic variables such as oil prices.

Another important observation from the forecasts is the increase in uncertainty as the forecast horizon becomes longer. This is reflected in the widening confidence intervals in the ARIMA forecasts. Such behaviour is expected in time series forecasting because prediction errors accumulate over time.

Despite the advantages of the models us ed in this study, several limitations should be acknowledged. The analysis was based solely on historical oil price data without incorporating additional variables that may influence price movements. Factors such as global economic conditions, supply disruptions, and geopolitical conflicts can significantly impact oil markets. Including such external variables could potentially improve forecasting performance.

8. Future Work and Improvements:

There are several ways in which this analysis could be improved in future work. One possible improvement would be to incorporate additional explanatory variables that influence oil prices, such as global oil demand, inflation rates, or economic growth indicators. Including these variables could help capture broader economic dynamics that affect oil price movements.

Another potential improvement would be to experiment with more advanced deep learning architectures. Models such as Gated Recurrent Units (GRU) or transformer-based time series

models have shown promising results in recent forecasting studies and may provide improved predictive performance.

Furthermore, ensemble approaches that combine predictions from multiple models could be explored. By integrating the strengths of different modelling techniques, ensemble methods[4] often produce more robust and accurate forecasts.

9. Conclusion:

This study looked at both traditional and deep learning methods for oil price forecasting. The Long Short-Term Memory (LSTM) neural network and the ARIMA model were the two models used.

The findings demonstrated that both models could accurately depict the overall pattern of the oil price time series. However, in terms of error metrics and its capacity to track data oscillations, the LSTM model showed superior predicting performance. This demonstrates the potential benefits of deep learning algorithms for handling nonlinear and complex time series data.

Overall, the results imply that integrating cutting-edge machine learning approaches with conventional statistical methods can yield insightful information for time series forecasting. While no model can perfectly predict oil price movements, these approaches can help improve forecasting accuracy and support better decision-making.

Statistical models like ARIMA are still useful because of their interpretability and cheaper computational cost, even if the LSTM model performed better in this study in terms of predicted accuracy.

10. References:

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